How Artificial Intelligence Improves Decision-Making in Small and Medium-Sized Enterprises

Abstract

This article explores the transformative role of Artificial Intelligence (AI) in enhancing decision-making processes within small and medium-sized enterprises (SMEs). Through an analysis of recent literature and empirical studies, the article identifies key benefits, implementation challenges, and necessary organizational capabilities for effective AI adoption. Insights are drawn from various sectors to provide a comprehensive understanding of how SMEs can leverage AI to drive better strategy, risk management, and market adaptability.

Introduction

The introduction of Artificial Intelligence (AI) into the operational and strategic frameworks of small and medium-sized enterprises (SMEs) marks a significant shift in decision-making paradigms. AI technologies can analyze vast amounts of data rapidly, enabling firms to make more informed choices. However, the successful integration of AI into decision-making processes is not without challenges. This article synthesizes findings from recent studies to understand the implications of AI for SMEs, including managerial attitudes, implementation barriers, and the necessary organizational resources for effective utilization.

The Impact of AI on Strategic Decision-Making

Aiudi et al. (2026) highlight the critical ways in which AI enhances strategic decision-making within SMEs. Their case studies on six Italian manufacturing companies reveal that AI not only improves decision quality but also accelerates the process and increases accuracy. By decentralizing decision-making and enhancing communication, AI facilitates a more agile response to market changes. However, challenges such as financial constraints and skill shortages remain prevalent, inhibiting some SMEs from fully leveraging AI's capabilities.

Managerial Attitudes and Adoption Challenges

Pozzo et al. (2024) conducted a study among 83 Colombian SMEs, finding that managerial attitudes significantly influence the adoption of AI technologies in decision-making processes. Factors such as perceived ease of use, external support, and personal concerns play vital roles in shaping these attitudes. Their findings underscore the importance of fostering a supportive culture and addressing potential resistances to facilitate AI adoption, which can ultimately enhance decision-making efficiency.

Implementation Barriers and Organizational Capabilities

Oldemeyer et al. (2024) provide a systematic review of implementation challenges faced by SMEs in integrating AI into their operations. The study categorizes 27 challenges, emphasizing crucial aspects such as data availability, infrastructure needs, and employee skill levels as foundational for successful AI adoption. The OECD (2021) supports this perspective, noting that AI can yield predictive insights and better forecasts but requires SMEs to overcome barriers like a lack of data culture and financing constraints.

The Role of AI in Risk Management and Market Adaptability

The OECD (2025) outlines how SMEs can leverage AI for better risk management and market adaptability. By utilizing predictive analytics and real-time data, SMEs can effectively forecast sales trends, optimize budget allocations, and anticipate changes in consumer demand. However, the report emphasizes that SMEs need to develop suitable data management practices and invest in necessary technologies and skills to reap these benefits. This highlights that while AI has the potential to revolutionize decision-making, the specific conditions for its successful implementation must first be met.

Conclusion

In conclusion, Artificial Intelligence presents a powerful tool for enhancing decision-making in small and medium-sized enterprises. While the benefits are substantial—ranging from improved accuracy and speed to better risk management—SMEs must navigate various challenges, including managerial attitudes, implementation barriers, and the need for organizational capabilities. Developing a supportive environment that fosters AI adoption and addressing these challenges will be instrumental in unlocking the full potential of AI for SMEs.

References

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